

Components of Food

Nutrients

The chemical substances in the food that are needed for the growth and development of the body.

Nutrition

The intake and utilisation of nutrients present in the food for the growth and development of the organism.

Q. What are the *major nutrients* present in our food?

Ans.

- > Carbohydrates
- > Fats
- > Proteins
- > Vitamins
- > Minerals

Also the following are required

- > Water
- > Roughage or dietary fibres

Food Groups

Classified into

1. Energy-giving food
2. Bodybuilding food
3. Protective food

Components of Food

Carbohydrates

1. Primary source of energy.
2. An organic compound made up of
 - i. Carbon
 - ii. Hydrogen
 - iii. Oxygen

Q. What are the *main forms* and *rich sources* of carbohydrates?

Ans. Main forms -

Starch, cellulose, and sugars

Rich Sources -

Cereals, sweet potato, potatoes, dairy products, tapioca, and corn.

Types of carbohydrates

	Features / Factors	Simple carbohydrates / Simple Sugars	Complex carbohydrates
1	Made up of	One (1) or two (2) sugar molecules	Long chain of sugar molecules (glucose)
2	In cold water	Easily soluble (Glucose)	Insoluble (Starch)
3	Taste	Sweet (Glucose)	Tasteless (Starch)
4	Digestion time	Quick (instant source of energy)	Long (release energy slowly)
5	Examples	Glucose, Fructose, and Galactose	Starch

Q. What happens with the *extra glucose* in the body?

Ans. The cells in the *liver* convert the extra glucose into **glycogen**, which is stored in the liver and muscle cells for future use.

Common sources of simple sugars

Common sources	Simple sugars
Fruits	Fructose, sucrose
Honey	Sucrose, glucose, maltose, and fructose
Milk	Lactose
Table sugar	Sucrose
Jaggery	Rich source of - sucrose Small quantities of - Glucose and fructose

Starch

Tasteless, odourless white substance

Q. What are the main sources of starch?

Ans.

Rice, wheat, maize, and potatoes.

Fats

Act as -

-> A Storehouse of energy in the body.

-> An insulator, stored under the skin to help maintain the body temperature.

Classified on the **basis of - Source**

	Features / Factors	Vegetable fats	Animal fats
1	Obtained from	Plants	Animals
2	Type	Unsaturated fats	Saturated fats
3	State - room temperature	Liquid	Solid
4	Sources	Nuts Groundnut oil Coconut oil Sunflower oil	Dairy products like - Butter Cheese Ghee Lard (pork fat) Tallow (beef fat)

Q. How are fats protective for the body?

Ans. Fats act as a cushion for the protection of internal organs from damage.

Example -  heart, and  kidney

Proteins

Are the building blocks of the human body.

Required for

-> Growth

-> Repair of tissues

- Worn-out or
- Damaged

Q. What are enzymes?

Ans. **Enzymes** are some **kind of proteins** that help in **controlling** the **activities** inside the body.

Sources

Plants – Pulses, soybean, and nuts.

Animals – Meat, fish, eggs, whey, and milk.

Functions of protein

Q. How is **haemoglobin** helpful?

Ans.

-> Haemoglobin is a **protein**.

-> Our blood contains haemoglobin that *transports / carries* **Oxygen** to different organs of the body.

Q. What are **enzymes** and how are they helpful?

Ans.

-> Most of the chemical reactions in the body use enzymes.

-> Enzymes are proteins that  **Increase** the **speed** of the **chemical reaction** and

-> Help in completion of the reaction

Vitamins -

Required in small quantities for the maintenance of our body

Types of Vitamins –

	Features / Factors	Fat-soluble	Water-soluble
1	In water	<i>Insoluble</i>	<i>Soluble</i>
2	Vitamins	A, D, E, and K	B₁, B₂, B₃, B₅, B₆, B₇, B₉, and B₁₂
3	Stored	In the body along with <i>fat tissues</i>	Cannot be stored Q. Why do we need to have them regularly? Ans. They are - -> Absorbed through the blood. -> Excess amounts are thrown out of body. -> Hence, need to be taken regularly in our diet.

Minerals

-> Required in small amounts.

-> For proper functioning of our body.

Example - calcium, iron, iodine, phosphorus, sodium, etc.

Classified on the basis of our **needs**

	Features / Factors	Macro minerals	Micro minerals / Trace minerals
1	Required in	Large amounts	Very small quantities
2	Example	Calcium, Phosphorus Potassium Chlorine Sodium Magnesium Sulphur	Copper Zinc Molybdenum Fluorine Manganese

Roughage (Dietary fibre)

Defined as **indigestible substance** which constitutes mainly of *Cellulose*.

Sources

Plant-based foods like vegetables – cereals, pulses, legumes, and nuts.

Water

Is a transparent, colourless and odourless liquid.

Functions

About 70% of the human body is made up of water.

- > Helps in maintaining the body temperature
- > Helps in transportation and absorption of nutrients
- > Helps in keeping us hydrated
- > Helps in the excretion of waste
- > Acts as a medium for the chemical reactions

Dehydration

Q. What is dehydration?

Ans. The condition in which a person **loses an excessive amount of water** from the body.

Balanced Diet

A diet that contains all the **essential nutrients** in the **right quantities**.

Q. What *factors* determine the *quantity of nutrients* in the diet?

Ans.

- > Age - Old person require less proteins in the diet
- > Workstyle - Athletes require more carbohydrates

Malnutrition

Intake of **less** or **excess** of **nutrients** in the diet.

Deficiency diseases / Nutritional disorder

The diseases that occur due to **lack of nutrients** over a long period of time.

Protein- Energy-Malnutrition (PEM)

- > Is a form of malnutrition
- > Caused due to deficiency of proteins or carbohydrates in the diet

Proteins	Carbohydrates
✓	✗
✗	✓
✓	✓

Symptoms of

		
	Kwashiorkor	Marasmus
1	Stunted growth	Loss of tissue and fat
2	Change in skin and hair colour	Sunken eyes
3	Dehydration	Chronic diarrhoea
4	Mental retardation	Weight loss
5	Bloated belly	Shrinkage of stomach

Mineral deficiency diseases

Caused due to deficiency of different minerals.